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Lab1: Robot Manipulation Tutorial: Stäubli Robotics Suite 2019 and VAL3

Task Description

"Pick part from A, place in B."

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Program Breakdown

Motion sequence (MoveJ, MoveL, etc.), I/O operations, waits.
Flow diagram or pseudocode.

Algorithm 1 Pseudocode with caption

```
Initialize robot and enable power
Set motion speed
Move to first pick position (Position 2a)
Close gripper to grasp object
Move to second position (Position 2b)
Close gripper again to secure object
Move to home position
Open gripper to release object
Display completion message
Reset motion and stop program
```

Task Parameters

The parameters used in this task include robot speeds, joint positions, Cartesian coordinates, tool frames, and payload settings.

Joint Positions

Table 1: Joint configuration for position0

J1	J2	J3	J4	J5	J6	J7
0	0	0	0	0	0	0

Cartesian Points

Table 2: Cartesian positions for pointRX (position2)

Index	X	Y	Z	Rx	Ry	Rz	Shoulder	Elbow	Wrist
0	379.5	-289.5	-285	-180	0	90	SFREE	EFREE	WFREE
1	547.0	-61.0	-265.18	-180	0	-90	SFREE	EFREE	WFREE

Tool Frame

Table 3: Tool configuration (eina0)

Index	X	Y	Z	Rx	Ry	Rz	Otime	Ctime
0	30	30	170	-90	44	-90	1	1
1	-30	-30	175	0	0	-134	1	1

Robot Speed

The nominal speed of the robot is set to $mNomSpeed.vel = 30$, which determines the velocity for joint and linear movements during the execution of the program.

Safety Considerations

The robot operates at high speeds and can be dangerous. Collisions are avoided by using the simulation environment to check trajectories before execution and enabled stop on first collision option. Positions such as `position0` serve as a safe home position, ensuring the robot starts and ends in a secure configuration. Tools and points are carefully defined, and the controller's limits and safety interlocks prevent unintended movements.

Execution Results

The robot successfully completed the pick-and-place sequence. Total cycle time was approximately 22:35 seconds. The robot performed 3 main movements (MoveJ to position2[0], MoveJ to position2[1], MoveJ back to position0) and operated the tool twice. No collisions were detected, and all I/O actions were executed as expected.

Improvements

The task could be optimized by:
1. **Reducing cycle time**: Adjusting joint speeds or using linear (MoveL) motions only when precise placement is required.
2. **Minimizing robot movements**: Reordering pick-and-place sequences to reduce unnecessary travel.
3. **Tool optimization**: Using a more suitable gripper or adjusting tool positions to reduce opening/closing time.
4. **Collision safety**: Fine-tuning points and trajectories to increase clearance and reduce collision risk.
5. **Automation**: Adding sensors to detect the presence of parts and dynamically adjust positions, reducing idle time.

Equipment setup

Image or diagram showcasing the setup for the application execution.



(a) Setup1 (b) Setup2

Figure 1: Application setup.